LAB EXAM — PART 2

# Customer Churn Prediction for an E-commerce Platform

A propensity model to identify at-risk customers and prioritize retention actions


STUDENT

Marta Paniconi · ID 535358


COURSE

AI for Communication and Marketing

PROGRAM

B.Sc. in Artificial Intelligence — A.Y. 2025/2026

# Predictive Model & Top Churn Risk Drivers



**5,626**
CUSTOMERS ANALYZED

**16.8%**
CHURN RATE (IMBALANCED)

**87.3%**
RECALL ON CHURNERS (RF)

**5.6×**
LIFT ONTOP-10%RISK DECILE

### Model Comparison — why Random Forest wins

| MODEL | ACCURACY | PRECISION | RECALL | F1 | AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.793 | 0.440 | 0.838 | 0.577 | 0.884 |
| ★ **Random Forest** | **0.928** | **0.745** | **0.873** | **0.804** | **0.976** |

*Recall is the key metric: **missing a churner is costlier than a false alarm.** RF keeps recall high (87%) while reducing false positives from 303 → 85.*

### Confusion Matrix — Random Forest (test set, n=1,688)

| Actual \ Predicted | Retained | Churn |
|---|---|---|
| Retained | 1,319<br>True Negative | 85<br>False Positive |
| Churn | 36<br>False Negative | 248<br>True Positive |

### TOP 3 CHURN RISK DRIVERS

Random Forest feature importance, reordered by business impact:

1 **Early lifecycle risk — Tenure**
Importance **30.1%**. Customers in their first **0–3 months** churn at **41.9%** — vs only **6–10%** for 4–24 month tenure.

2 **Negative experience — Complaints**
Importance **7.9%**. Customers who complained churn at **31.7%** vs **10.9%** — **almost 3× higher**. Strongest experience signal in the data.

3 **Incentive sensitivity — Cashback & Activity**
**CashbackAmount** (8.5%) and **DaySinceLastOrder** (5.6%) reveal value-engagement patterns. Long gaps + low cashback = elevated risk.

**Class imbalance handled** via `class_weight="balanced"` + stratified 70/30 split. Recall prioritized over precision.

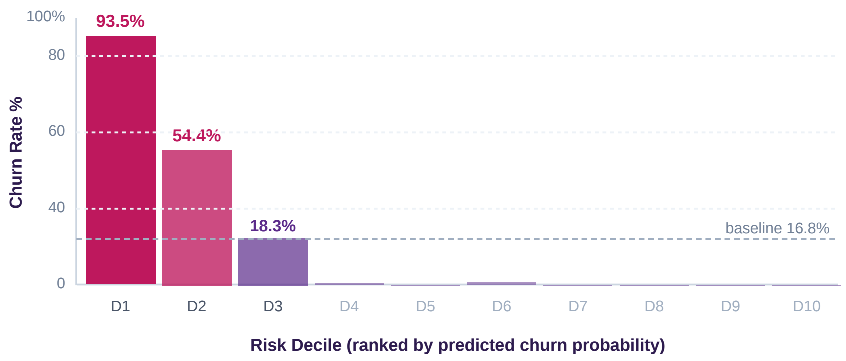


Marta Paniconi · 535358

# Strategic Recommendation — Targeted Retention Campaign



## Lift Chart — top deciles concentrate the churners




**Marketing efficiency:** targeting only the top 20% (D1+D2) captures **~74%** of all churners. A blanket campaign would waste 80% of the budget.

### WHY RECALL > PRECISION HERE

Acquiring a new customer costs **5–7×** more than retaining one. A false positive = a small incentive sent to a loyal customer (low cost). A false negative = a lost customer (high cost). **Optimizing for recall is the right business choice.**

RECOMMENDED CAMPAIGN

## "Welcome & Win-Back" Two-Wave Retention

*Target the top **2 deciles** (~20% of customers) flagged by the model. Two waves, differentiated by root cause of risk.*

### WAVE 1 — EARLY-LIFECYCLE ONBOARDING

**Who:** high-risk customers with Tenure < 4 months.

**How:** proactive customer-service outreach + a personalized welcome coupon on their preferred category (Mobile, Laptop & Accessory).

**Why:** the 0–3 month band concentrates **41.9%** churn rate — habit-building is the priority.

### WAVE 2 — COMPLAINT RECOVERY

**Who:** high-risk customers flagged by Complain = 1.

**How:** apology email + dedicated service line + cashback boost + free shipping for next 2 orders.

**Why:** complainers churn at **3× the rate**. Service recovery converts at-risk customers into advocates.

**Expected outcome:** acting only on the top 20% (≈340 customers in test) addresses ~74% of likely churn with **1/5** of the budget of a full-base campaign.